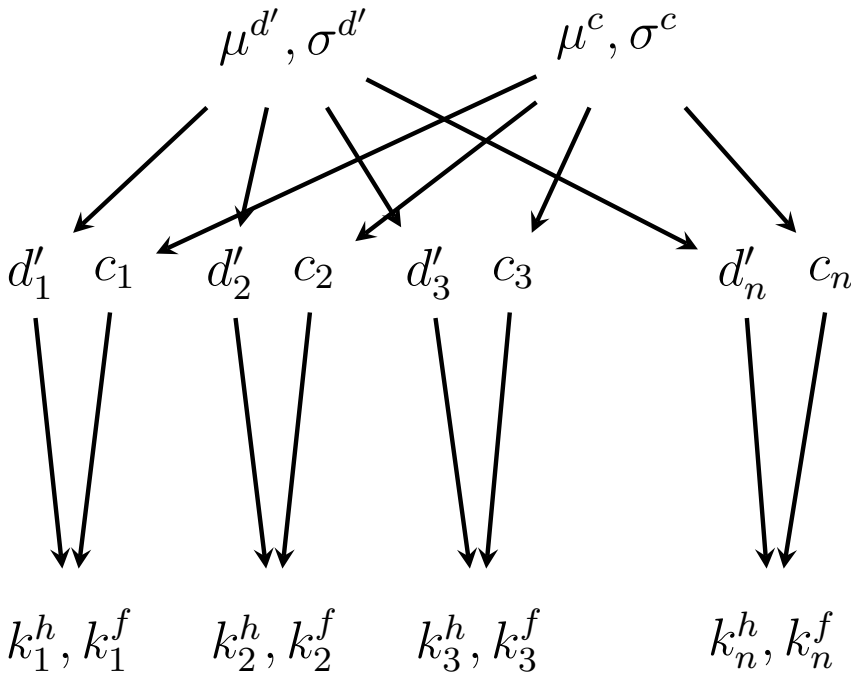
Hierarchical Two-Choice Signal Detection Model

Michael D. Lee

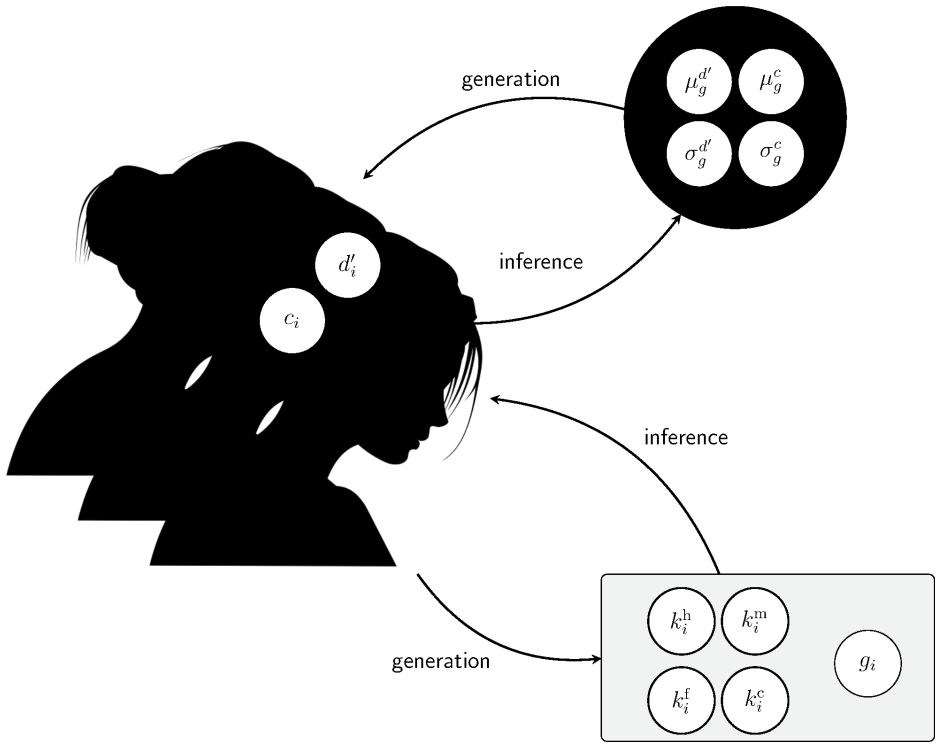
Motivation for Hierarchical Model

- The motivation for a hierarchical two-choice signal detection model is the same as for the rate model
  - allow for sameness and difference: allow for individual discriminability and bias to differ from each other, but belong to common group distributions
  - new inferences: the properties of the group distribution are inferred, as well as inferences about a representative individual



Two-Choice Signal Detection Model Structure

- The hierarchical two-choice signal detection model has the structure
  - observed counts of hits k_i^{h}, false alarms k_i^{f}, misses k_i^{m}, and correct rejections k_i^{c}
  - known allocation of each individual to a group g_i
  - individual discriminability d_i' and bias c_i parameters
  - the discriminabilities and biases come from Gaussian group distributions with means $\mu_g^{d'}$, μ_g^c and standard deviations $\sigma_g^{d'}$, σ_g^c



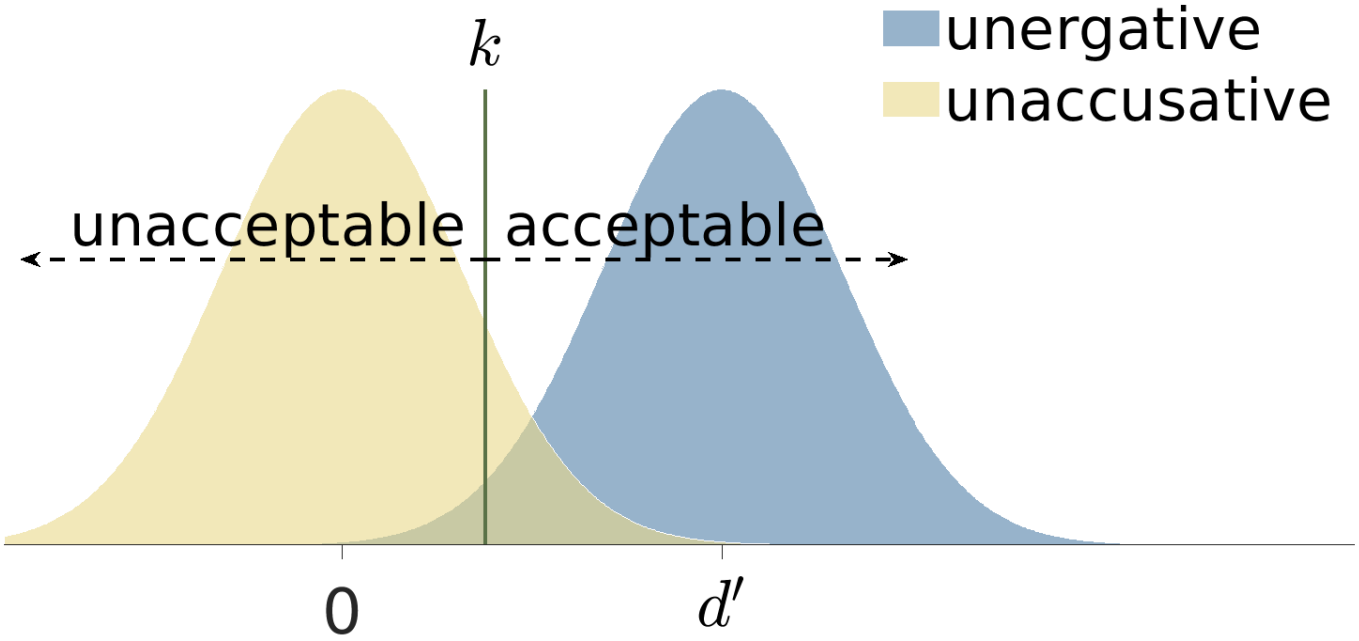
Grammatical Acceptability

The Unaccusative Hypothesis for –er Word Acceptability

- Huang and Ferreira (2020) studied how people make acceptability judgments about the –er nominalization of verbs
  - e.g., adding "er" to the verb "run" to get "runner" seems acceptable, but adding "er" to "arrive" to get "arriver" does not
- The experiment asked 20 native English speakers to judge the acceptability of 60 –er words
  - 30 words based on unergative (agent) verbs like "run", "dance", "hop", "jog"
  - 30 words based on unaccusative (theme) verbs like "arrive", "melt", "sink", "fall"
- The Unaccusative Hypothesis predicts that only the unergative verbs will lead to acceptable –er words

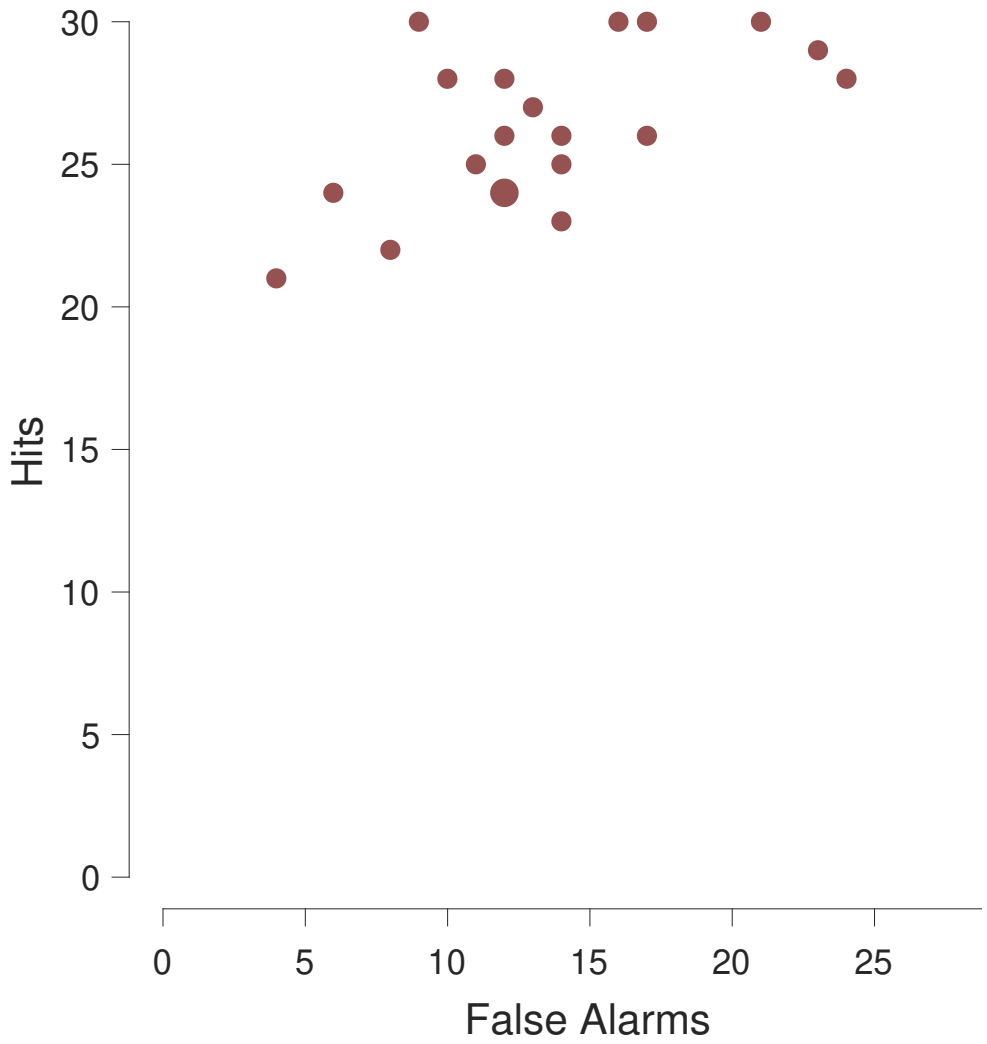
Signal Detection Theory Framework

- Following the Unaccusative Hypothesis, Huang and Ferreira (2020) use signal detection theory by
  - treating unergative verbs as signal stimuli and unaccusative verbs as noise stimuli
  - the criterion determinesing whether a participant's judgment is acceptable or unacceptable



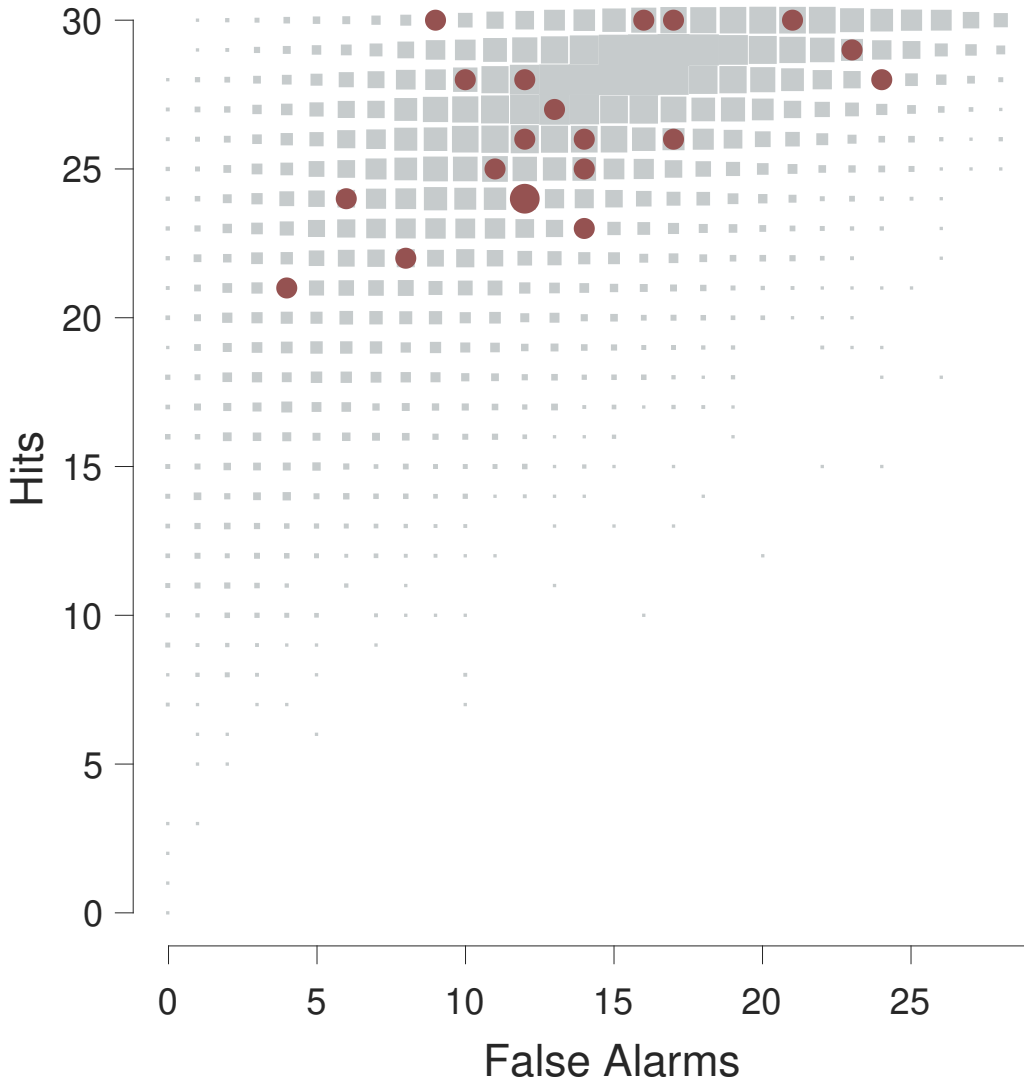
Data

- The hit and false alarm counts the participants are shown below



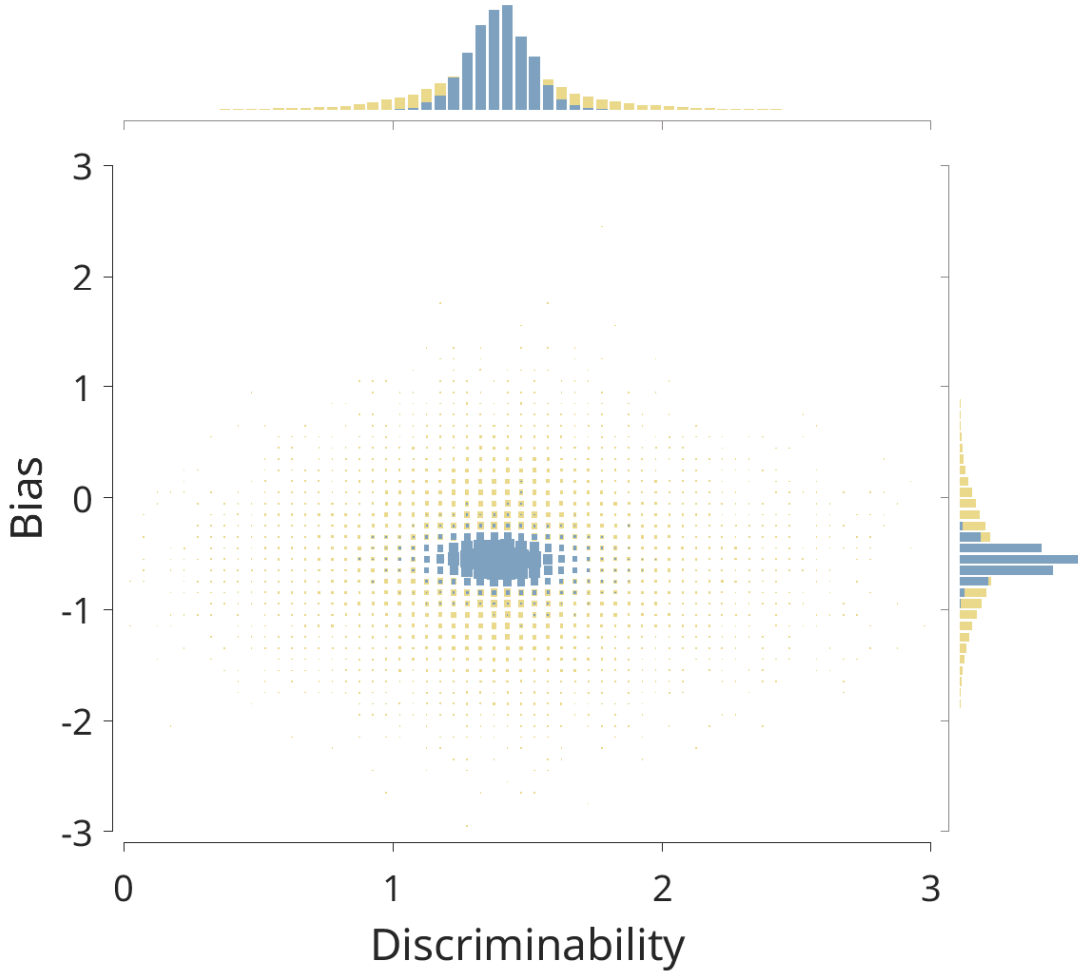
Descriptive Adequacy

- The squares represent the posterior predictive distribution for the representative participant, and are overlaid on the individual data



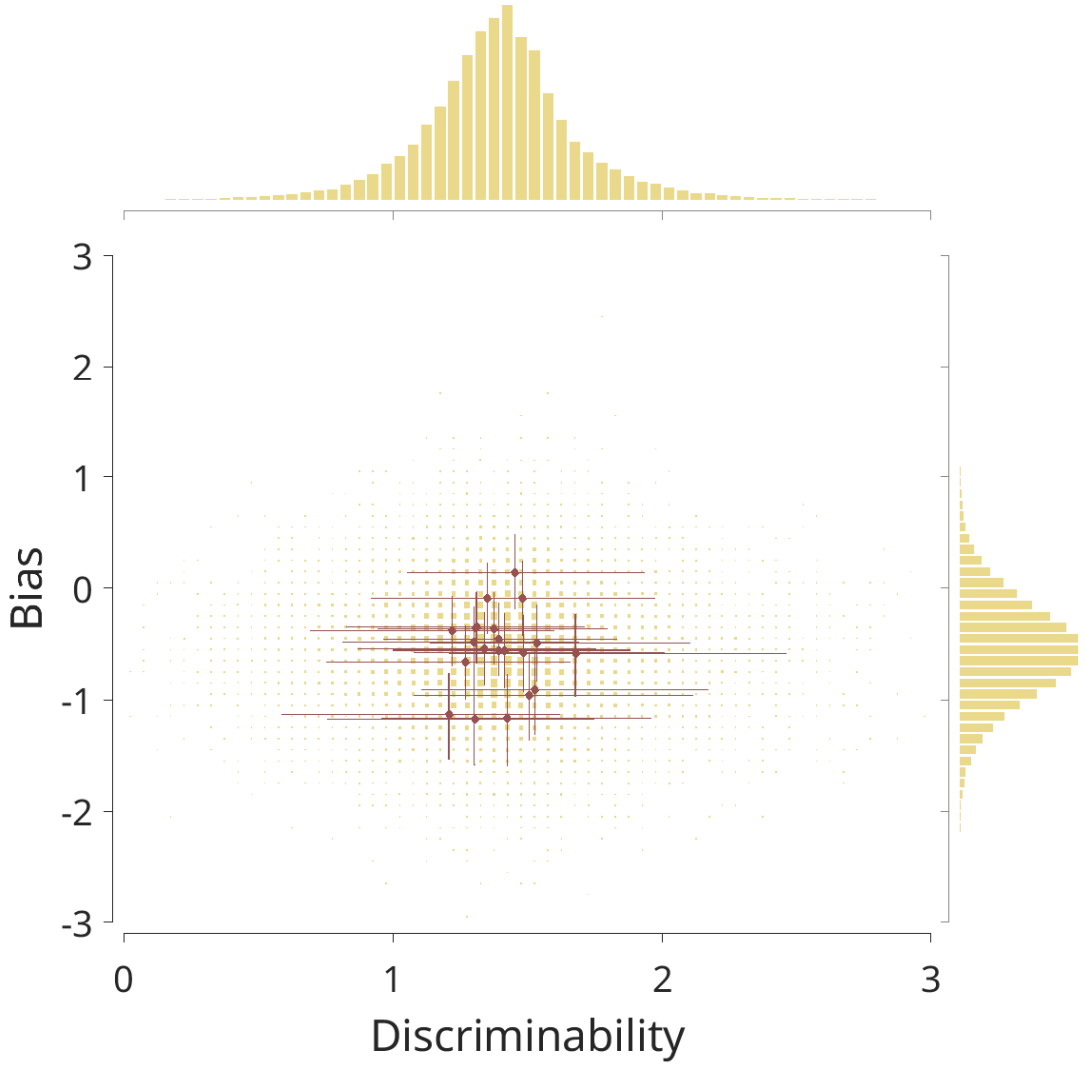
Group-level Inferences

- The joint and marginal posterior distribution are shown for
  - the group means $\mu_{d'}$ and μ_c in blue
  - the representative individual $\tilde{d}'$ and $\tilde{c}$ in yellow



Individual-level Inferences

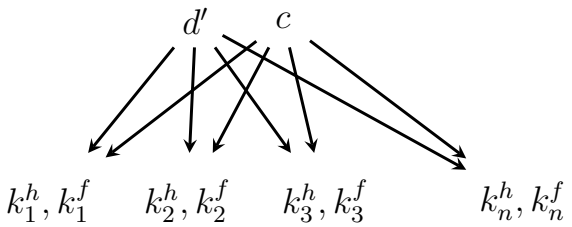
- The red points correspond to experimental participants and the yellow distributions correspond to the representative participant



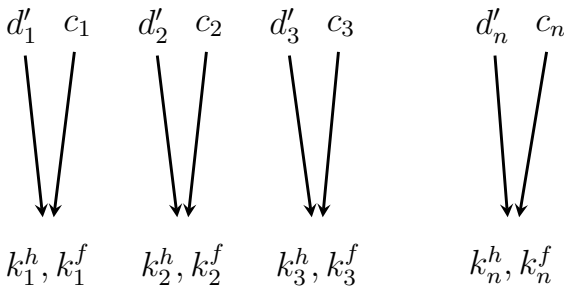
Huang and Ferreira (2020) Signal Detection Theory Analyses

- Huang and Ferreira (2020) report two sorts of SDT analysis
  - the first sums the hits and false alarms over participants for one overall d' and c estimate
    - this corresponds to assuming every participant has exactly the same d' and c
  - the second estimates d'_i and c_i for each participant based on only their hit and false alarm counts
    - this corresponds to assuming every participant has a d'_i and c_i that is completely independent of every other participant

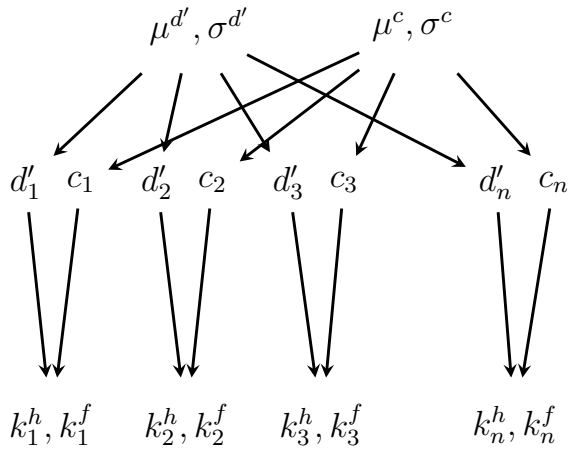
Same



Independent

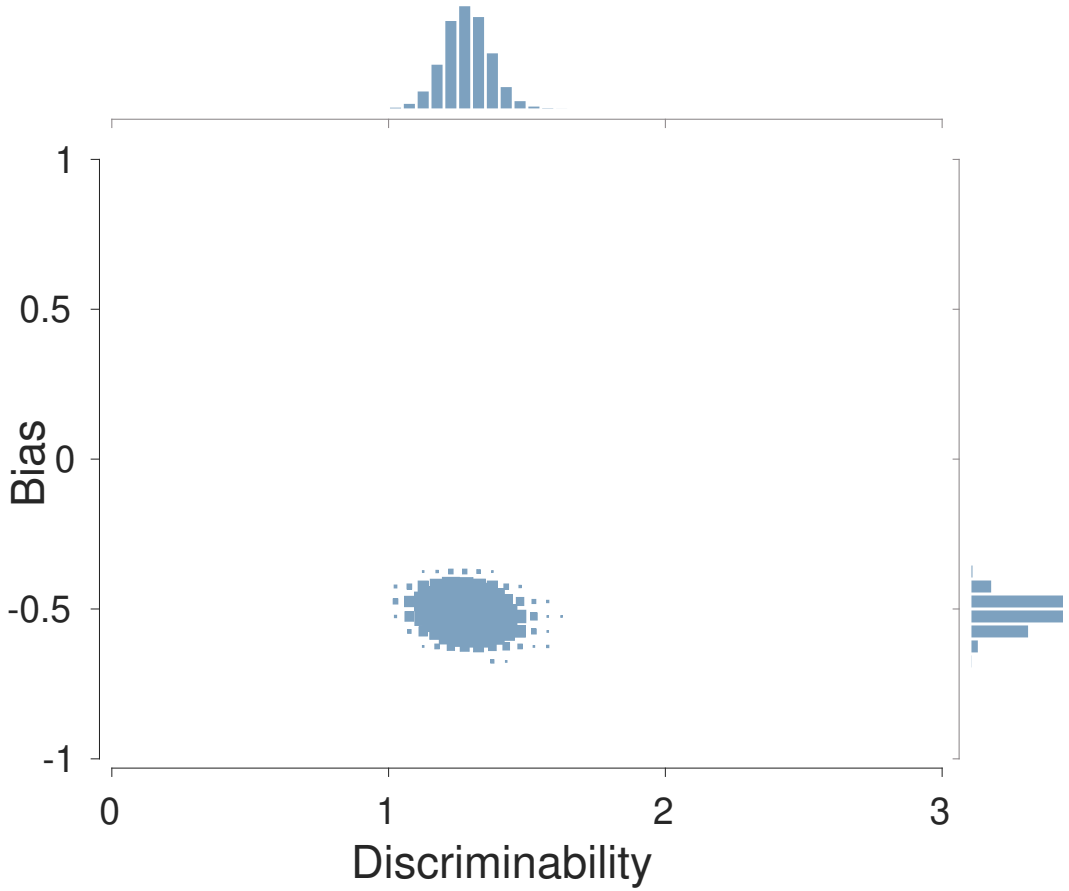


Hierarchical



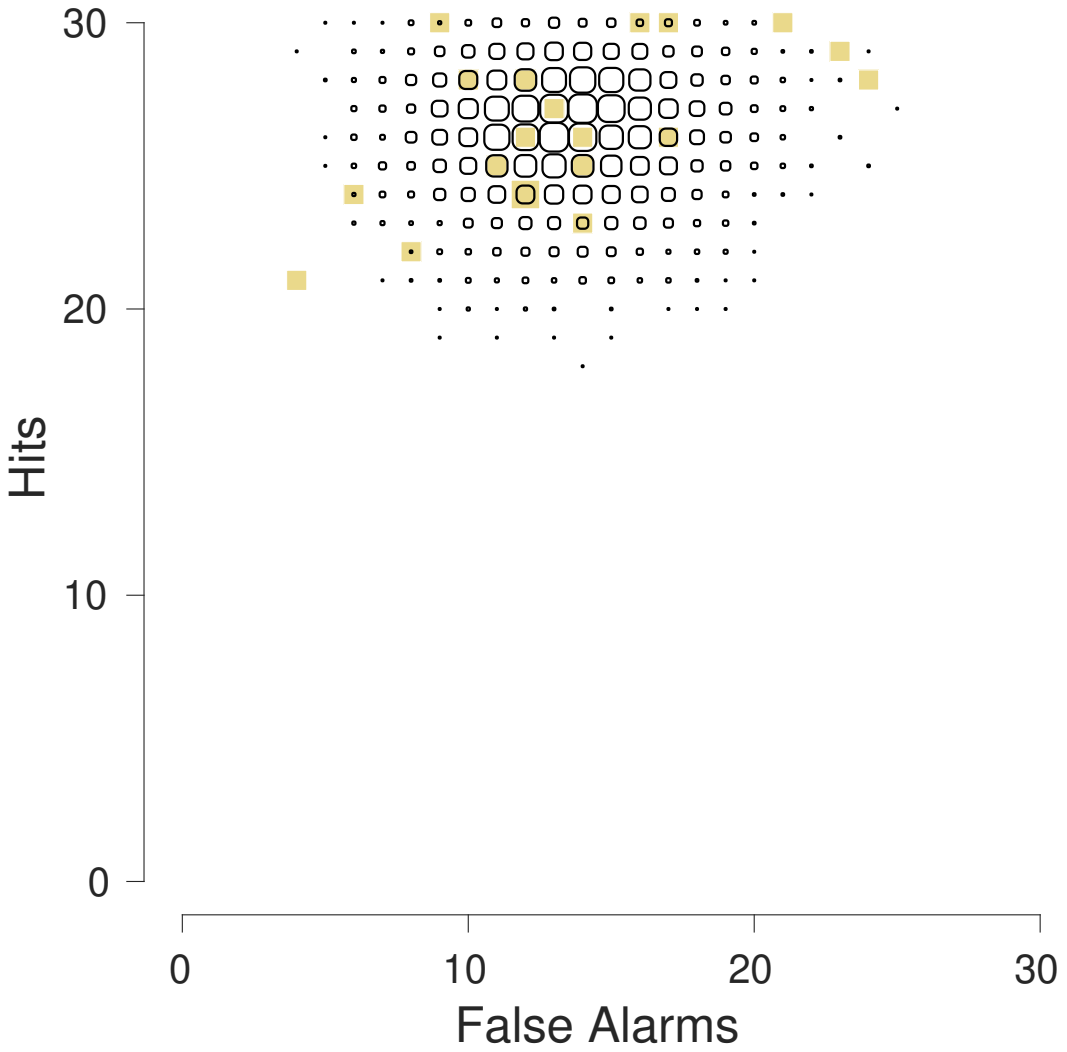
Inferences of "Same" Model

- Bayesian inferences from the non-hierarchical two-choice signal detection model that assumes the same d' and c are consistent with Huang and Ferreira's (2020) frequentist estimates of $d' = 1.288$ and $c = -0.514$



Descriptive Adequacy of "Same" Model

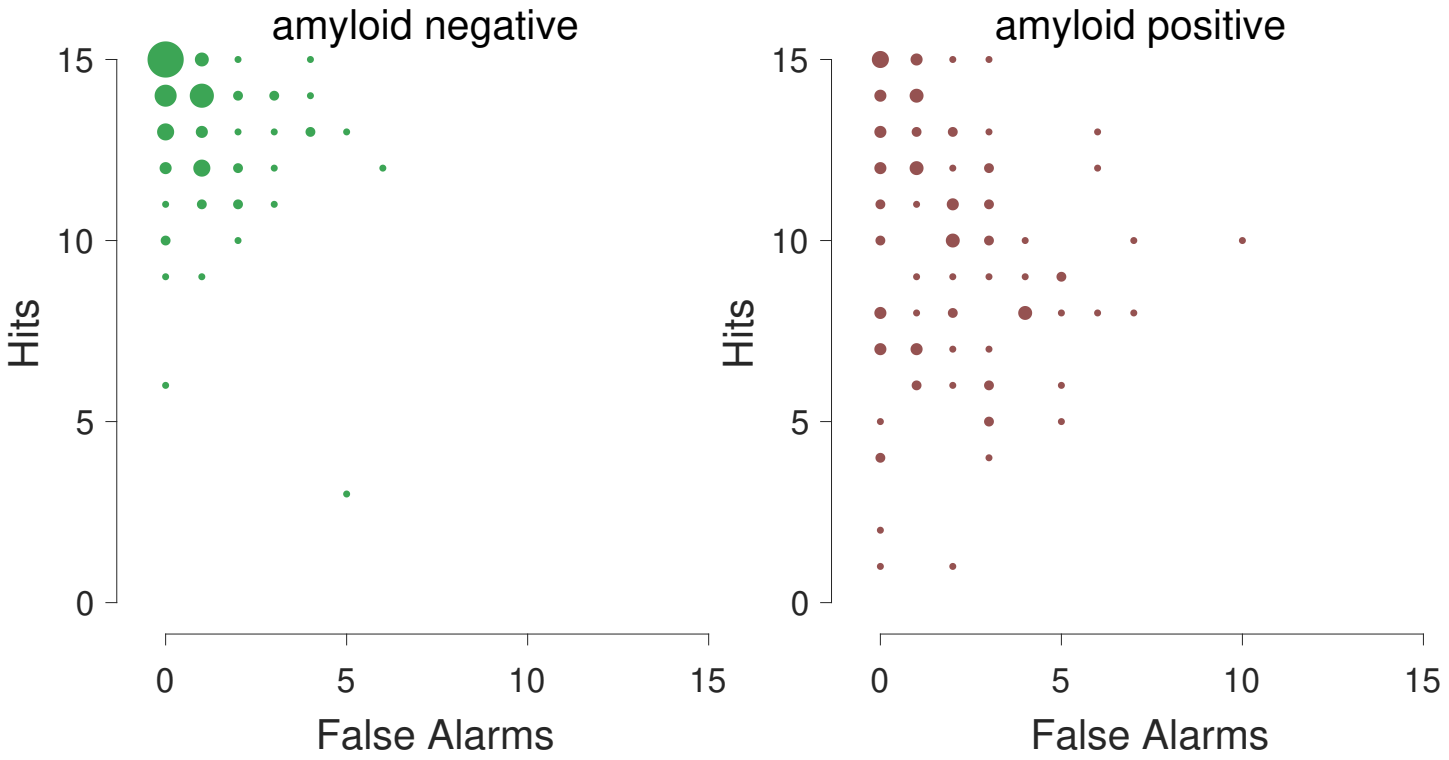
- The squares represent the posterior predictive distribution for the representative participant, and are overlaid on the individual data



Amyloid Memory

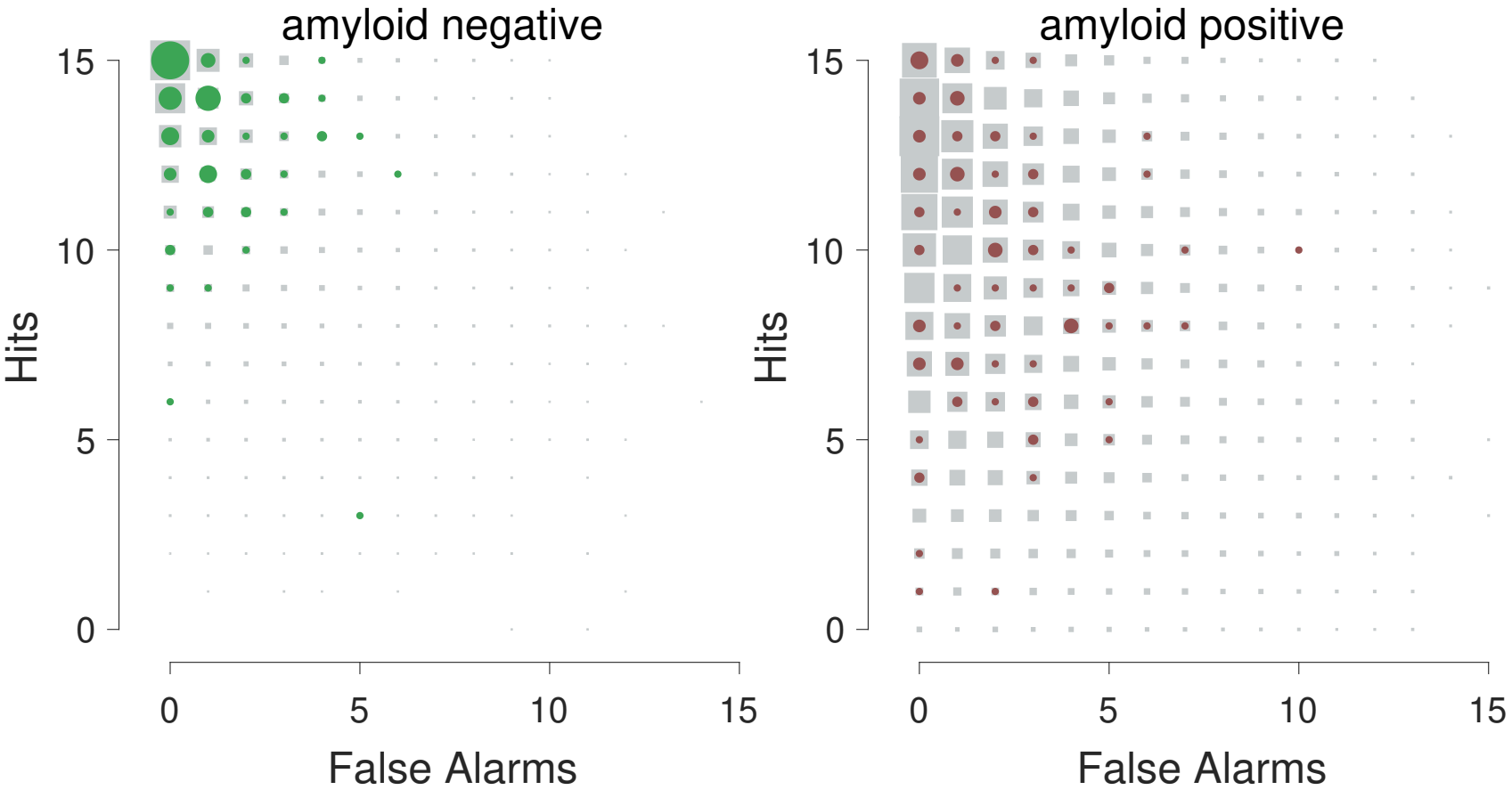
Recognition Memory and Amyloid

- 200 patients aged over 75 years were clinically assessed
  - memory: via the RALVT recognition memory task (Bean, 2011), with 15 old words and 15 new words
  - amyloid: classified as amyloid positive or negative, based on a cerebrospinal fluid measurement
- The hit and false alarm counts for each group are shown



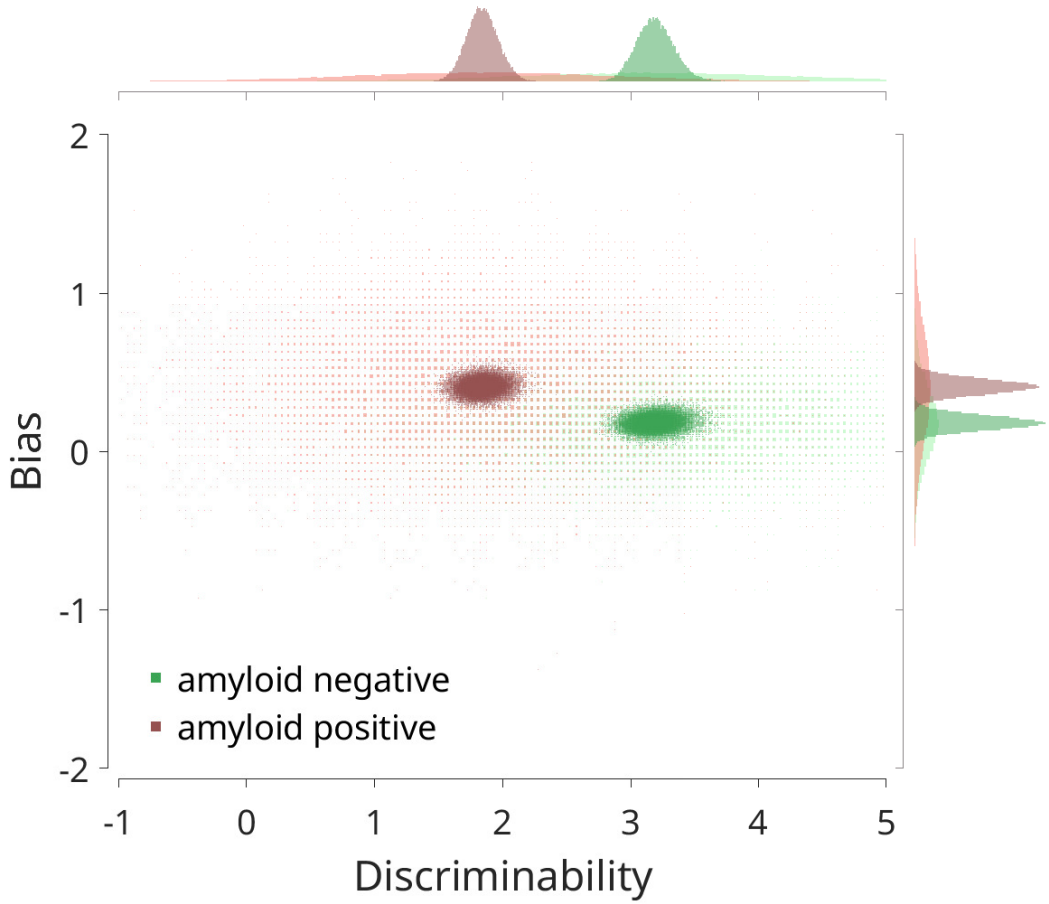
Descriptive Adequacy

- The squares represent the posterior predictive distribution for the representative participant, and are overlaid on the individual data



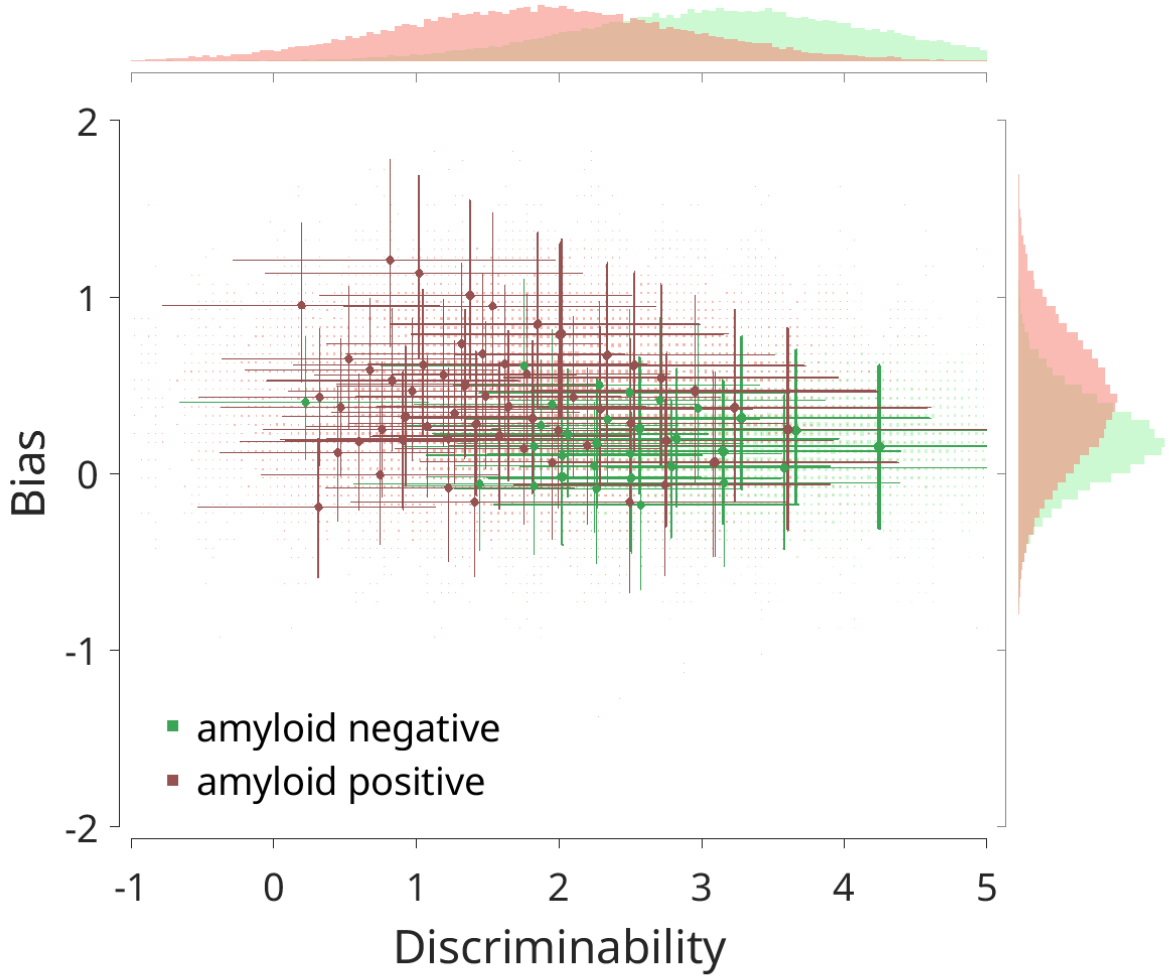
Group-level Inferences

- The joint and marginal posterior distribution for the group means $\mu_{d'}$ and μ_c are shown
  - any change in decision strategy is minor'
  - amyloid positive group has much worse memory



Individual-level Inferences

- The representative and individual participant posterior distribution show there is still some overlap at the individual level
  - differences at the group and individual level mean different things



References i

Bean, J. (2011). Rey auditory verbal learning test. *Encyclopedia of Clinical Neuropsychology*, 2174–2175.

Huang, Y., & Ferreira, F. (2020). The application of signal detection theory to acceptability judgments. *Frontiers in Psychology*, *11*, 73. doi: 10.3389/fpsyg.2020.00073